

## RESEARCH ARTICLE

# Design and Implementation of Multiple Ring Oscillator-Based TRNG Architecture by Using ADPLL

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**ABSTRACT** A new technique for generating true random numbers by using the ADPLL (All Digital Phase Locked Loop)-based multiple ring oscillator TRNG (MURO-TRNG) is discussed in this paper. The proposed ADPLL-based MURO-TRNG contains 10 ring oscillators, 1 conventional ADPLL, 11 sampling DFFs, 1 XOR gate, and an XOR corrector-based post-processing circuit. Ring oscillators are the entropy sources for the proposed MURO-TRNG architecture, and they are constructed by ADPLL with different frequencies. A new DCO (Digital Controlled Oscillator) constructed by using 9 NOR gates and 1 DFF is designed for constructing ADPLL-based ring oscillator circuits. Conventional ADPLL operates at 3 different reference frequencies to sample the raw random bits and to provide a clock for the post-processing circuit. The proposed MURO-TRNG architecture is designed using VHDL, implemented on the Artix 7, Kintex-7, and Zynq7000 FPGAs, and simulated by the Xilinx Vivado 2015.2 tool. The designed and implemented MURO-TRNG architectures consume 2-4 LUTs and 2-4 FFs. Energy consumption per bit of the generated bitstream is in the range of 4.22 nJ/bit-5.85 nJ/bit, and throughput values are in the range of 206.82 Mbps-260.07 Mbps. The NIST SP 800-22 test is conducted to validate the randomness of the generated bit stream outputs from the post-processing circuit.

**INDEX TERMS** TRNG, DCO, ADPLL, MURO-TRNG, NOR, XOR corrector.

## I. INTRODUCTION

Random numbers generated are of long length, and they are used to provide keys for encrypting sensitive data. Once data is encrypted using a random number, it is not possible for another person or computer to decrypt the data. Random numbers are categorized into two sub-categories. 1. Pseudo-random numbers and 2. True-random numbers. Pseudo-random number generators use mathematical algorithms to generate random numbers, and they don't require any external inputs [1]. True randomness uses physical entities such as radioactive decay, entropy, or electrical noise to generate true randomness [2]. By utilizing various random differences in the physical process, TRNG (true random number generator) produces infinite random number sequences [3].

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In comparison with the pseudo-random number generator (PRNG), TRNG provides highly secure random, unpredictable, and unrepeatable sequences. TRNG is classified into the following categories: 1. Johnson noise-based TRNGs 2. Jitter-based TRNGs 3. Metastability-based TRNGs 4. Chaos-based TRNGs 5. Mixed-based TRNGs 6. Quantum-based TRNGs. Today, TRNG is used in many applications, such as cryptography, online gaming, scientific research, simulations, quality assurance, randomized controlled trials, security, art and music, and electronic warfare. For the practical aspect, it is highly desirable to implement TRNG using purely digital techniques. By using digital techniques, TRNG can be easily implemented on popular reconfigurable devices such as FPGAs and CPLDs. TRNG can be designed by using noise sources from analog PLLs (phase-locked loops). But analog PLL-based TRNG designs consume more area, less speed, less flexibility, and are more complex compared with TRNG

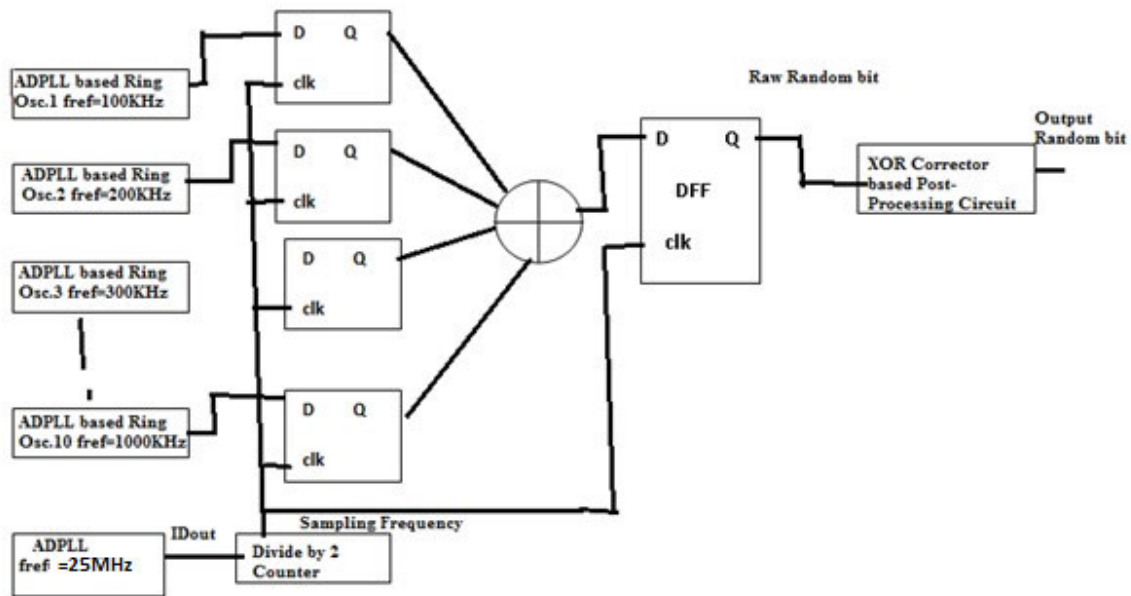


FIGURE 1. ADPLL based MURO-TRNG architecture.

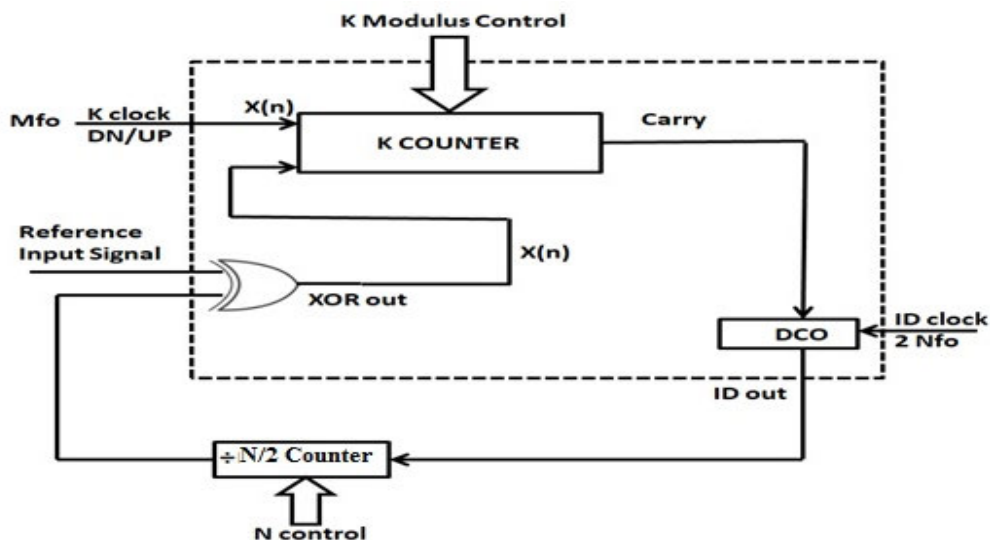
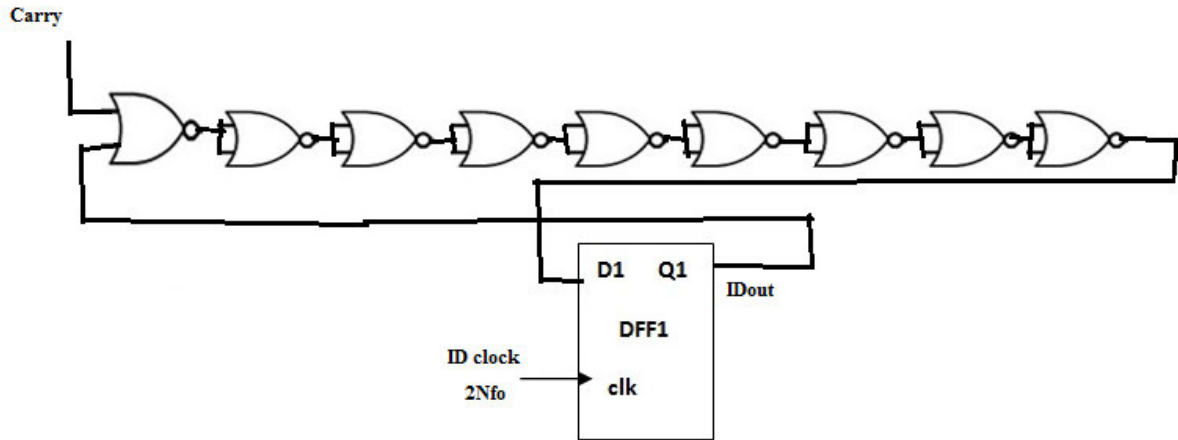


FIGURE 2. Proposed ADPLL block diagram.

based on FPGA Digital Logic architecture [4]. Fully digital PLLs named ADPLL consume less area, have higher speed, and are less complex [5]. For the last twenty-five years, many researchers have been working on designing TRNG architecture. Recently, the author in paper [6] designed TRNG architecture by using avalanche noise. This architecture uses two independent entropy channels to generate random bytes. An ATmega32U4 microcontroller with USB interface was used to implement this design, and it produces a throughput of 136.0 Kbit/s without using any post-processing algorithms. Generated random bits by this design are evaluated for bias and serial correlation and the NIST 800-22R1a test

is conducted to verify the randomness. Cao et al. [7] in 2023 proposed a high-energy efficient TRNG architecture using a current starve based ring oscillator (CSRO). In this design, CSRO is biased in the sub-threshold region, and it is used to sample ring oscillator (RO) to produce multichannel sequence output. 40nm CMOS technology was used to implement and simulate this architecture. This architecture consumes  $11.46 \mu\text{W}$  at  $1.1 \text{ V}$ ,  $25^\circ\text{C}$ , energy consumption per a bit for the generated bitstream is  $0.555 \text{ pJ/bit}$  and throughput value for this is  $20.66 \text{ Mbps}$ . By using open loop structure, authors in paper [8] designed very high speed TRNG architecture. As proposed architecture uses mere wire delay



**FIGURE 3.** Proposed DCO circuit diagram for designing ADPLL based ring oscillator circuits.

chain it is easily implemented on any FPGA or ASIC device. Proposed architecture was tested on the STRATIX FPGA and NIST test was conducted to verify the randomness of the generated bitstream. High speed TRNG architecture can be designed by combining phase jitter produce by ring oscillators with an XOR gate. Author in paper [9] designed a low-cost and high speed TRNG by mixing ring oscillators output with an XOR gate. Proposed TRNG was implemented on Artix7 FPGA (XC7A35T-1FTG256C) board and it consume 25LUTs, 37FFs and achieve throughput value of 160Mbps. NIST SP800- 22 and AIS31 test were conducted for evaluating the randomness of the generated bitstream. But in this design simulation output waveform was not shown. Manish and Bharti in 2021 [10] designed uses random jitter produced by free running oscillator to generate true random numbers. In this design, free running oscillators uses Programmable delay lines to add jitter and variations in the ROs. Proposed TRNG was implemented on Xilinx Artix-7 FPGA, and it consume 41 LUTs, 48 FFs and post-processing was conducted to remove the bias of random numbers. Simulation waveforms for the deigned TRNG with and without post-processing were shown in this paper. But power consumption and throughput results were not shown for this design. Several other researchers in the beginning phase of TRNG design have sampled phase jitter [11], [12], [13], [14], [15], [16] associated with ring oscillators to produce true random numbers. In this research, we deeply studied TRNG design based on ring oscillators, and we propose a new MURO- TRNG architecture using ADPLL. In our design, conventional ring oscillators are replaced by ADPLL, and they achieve low power consumption, low area consumption, high throughput. A new DCO (Digital Controlled Oscillator) based on NOR gates and DFF is proposed in building ADPLL based ring oscillators circuits. Conventional ADPLL operates at three different reference frequencies ( $f_0=25\text{MHz}$ ,  $12.5\text{MHz}$  and  $f_0=6.25\text{MHz}$ ) to generate three

different sampling frequencies. Our proposed design is synthesized and simulated using Xilinx Vivado 2015.2 tool and the simulation waveform clearly shows generated true random numbers. Proposed TRNG architectures are tested for voltage and Temperature variations using three FPGAs (Artix7, Kintex-7 and Zynq7000).

## II. PROPOSED ADPLL-BASED MURO-TRNG ARCHITECTURE

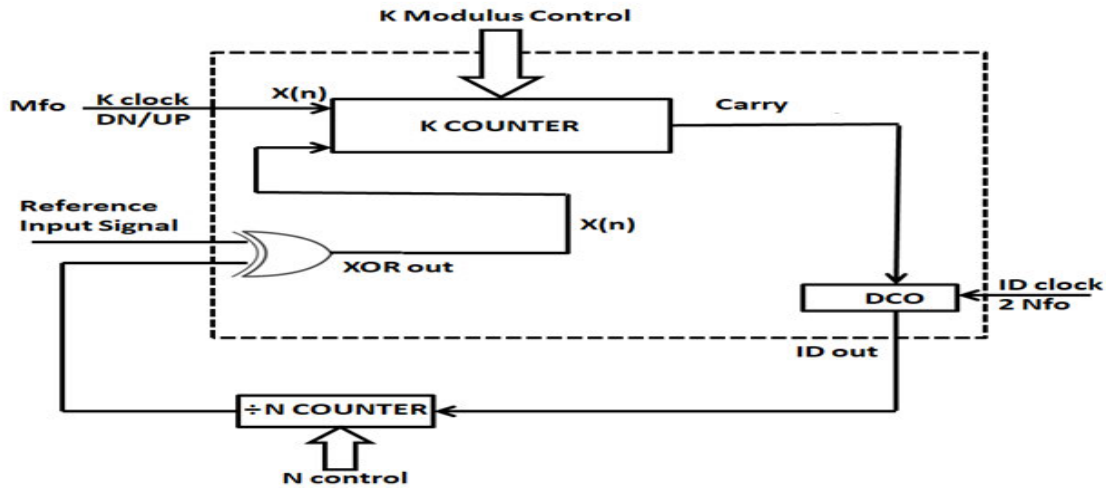
A new TRNG(True Random Number Generator) architecture based on multiple ring oscillator is shown in Figure.1. This architecture contains 10 ADPLL based ring oscillator circuits, 11 DFF based sampling circuits, 1 XOR gate, 1 Conventional ADPLL(with reference frequency  $f_0=25\text{MHz}$ ) with divide by2 counter circuit for generating sampling frequency and a XOR Corrector based post-processing circuit. ADPLL DCO outputs in the ring oscillators circuits are the entropy sources for the proposed MURO-TRNG architecture. Phase jitter produced by ADPLL DCO outputs in the ring oscillators circuits are sampled by DFFs before they XORed. MURO-TRNG architecture generates true random bits by using Phase Jitter. Sampling frequency for this architecture is generated by conventional ADPLL DCO output followed by a divide by 2 counter circuit. XOR gate in this circuit XORed ring oscillators' outputs with different clock frequencies. Output of XOR gate produces raw random bits. To reduce the bias from the generated raw random bits a XOR corrector based post-processing circuit is used in the proposed MURO-TRNG architecture. Detailed description of each components used in MURO-TRNG is discussed below.

### A. PROPOSED ADPLL BASED RING OSCILLATOR CIRCUIT

To produce true randomness from the MURO-TRNG architecture modified ADPLL based ring oscillators circuits are designed. Modified ADPLL contains, XOR gate based Phase

**TABLE 1.** Parameter for the ADPLL based ring oscillator circuits used in muro-trng.

| Parameter                                              | Ring Osc1 | Ring Osc2 | Ring Osc3 | Ring Osc4 | Ring Osc5 | Ring Osc6 | Ring Osc7 | Ring Osc8 | Ring Osc9 | Ring Osc10 |
|--------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|------------|
| K                                                      | 4         | 4         | 4         | 4         | 4         | 4         | 4         | 4         | 4         | 4          |
| M                                                      | 8         | 8         | 8         | 8         | 8         | 8         | 8         | 8         | 8         | 8          |
| N                                                      | 4         | 4         | 4         | 4         | 4         | 4         | 4         | 4         | 4         | 4          |
| Reference input signal( $f_o$ )                        | 100 KHz   | 200 KHz   | 300 KHz   | 400 KHz   | 500 KHz   | 600 KHz   | 700 KHz   | 800 KHz   | 900 KHz   | 1000 KHz   |
| K Counter clock( $Mf_o$ ) and DCO DFF clock( $2Nf_o$ ) | 800 KHz   | 1600 KHz  | 2400 KHz  | 3200 KHz  | 4000 KHz  | 4800 KHz  | 5600 KHz  | 6400 KHz  | 7200 KHz  | 8000 KHz   |


**FIGURE 4.** ADPLL block diagram for sampling frequency generation [21].

Detector circuit, K counter based Loop Filter circuit, NOR gates and DFF base a new DCO circuit and a divide by  $N/2$  counter circuit. Phase detector circuit [17], [18], [19] compare divide by  $N/2$  counter output with the ADPLL reference signal ( $f_o$ ) and when it outputs has logic low value then it trigger K counter. Frequency of K counter is M times the reference frequency ( $Mf_o$ ) of the ADPLL. MSB value of the K counter [20], [21] is used to trigger DCO circuit. DCO DFF clock is  $2Nf_o$ . Proposed ADPLL block diagram is shown in Figure.2. ADPLL DCO output is fed to divide by  $N/2$  counter. And then divide by  $N/2$  counter output is compared with reference signal to minimize the phase between reference and DCO. T FF is used to design divide by  $N/2$  counter circuit. Proposed DCO circuit diagram used in ADPLL is shown in Figure.3. It contains 9 two input NOR gates and 1 DFF. NOR gates used in the proposed DCO circuit uses constant rise time of value 10 ns. In this DCO, first NOR gate takes one input as MSB value of K counter circuit and other input is output of the DFF. In this figure, Carry represents MSB value of the K counter circuit and IDout represents DCO output which is fed into divide by  $N/2$  counter circuit in the ADPLL. Design parameter for the ADPLL based ring oscillator circuits is shown in Table 1. From this Table it is clear that ring oscillator1 uses 100 KHz as a reference signal for ADPLL

design, ring oscillator2 uses 200 KHz as a reference signal for ADPLL design, ring oscillator3 uses 300 KHz as a reference signal for ADPLL design, ring oscillator4 uses 400 KHz as a reference signal for ADPLL design, ring oscillator5 uses 500 KHz as a reference signal for ADPLL design, ring oscillator6 uses 600 KHz as a reference signal for ADPLL design, ring oscillator7 uses 700 KHz as a reference signal for ADPLL design, ring oscillator8 uses 800 KHz as a reference signal for ADPLL design, ring oscillator9 uses 900 KHz as a reference signal for ADPLL design, and ring oscillator10 uses 1000 KHz as a reference signal for ADPLL design. ADPLL DCO outputs with different frequencies of these ring oscillators circuits are used for generating timing jitter.

## B. SAMPLING FREQUENCY GENERATION FOR THE MURO-TRNG ARCHITECTURE

Conventional ADPLL DCO output with divide by 2 counter circuit is used for generating sampling frequency for 11 DFFs based sampling circuit in the proposed MURO-TRNG architecture. Conventional ADPLL and its DCO circuit diagram for the generating sampling frequency is shown in Figure.4 and 5 respectively. T FF is used to design divide by 2 counter circuit. Design parameters for the conventional ADPLLs

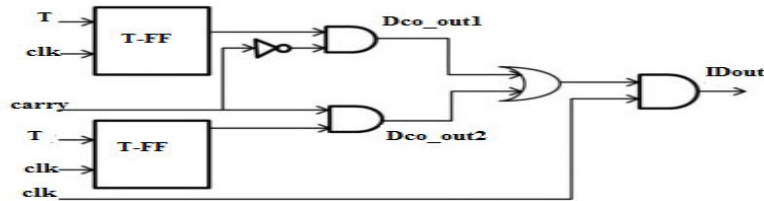


FIGURE 5. DCO circuit diagram [21].

(with three different reference frequencies) are shown in Table 2, 3 and 4 respectively. These ADPLLs work at 25MHz, 12.5 MHz and 6.25MHz reference input signals, use the XOR gate base phase detector circuit, a K counter circuit with clock frequency of value M times reference signal ( $Mf_0$ ), DCO circuit uses clock frequency of value  $2Nf_0$  and divide by N counter circuit. Logic low output value of phase detector circuit trigger K counter circuit to count in upward direction and MSB value of K counter circuit trigger DCO circuit in this ADPLL. K counter and DCO clock values are equal.

TABLE 2. Parameter for the ADPLL ( $f_0=25\text{MHz}$ ).

| Parameter                       | ADPLL design |
|---------------------------------|--------------|
| K                               | 4            |
| M                               | 16           |
| N                               | 8            |
| Reference input signal( $f_0$ ) | 25MHz        |
| K Counter clock( $Mf_0$ )       | 400MHz       |

TABLE 3. Parameter for the ADPLL ( $f_0=12.5\text{MHz}$ ).

| Parameter                       | ADPLL design |
|---------------------------------|--------------|
| K                               | 4            |
| M                               | 16           |
| N                               | 8            |
| Reference input signal( $f_0$ ) | 12.5MHz      |
| K Counter clock( $Mf_0$ )       | 200MHz       |

TABLE 4. Parameter for the ADPLL ( $f_0=6.25\text{MHz}$ ).

| Parameter                       | ADPLL design |
|---------------------------------|--------------|
| K                               | 4            |
| M                               | 16           |
| N                               | 8            |
| Reference input signal( $f_0$ ) | 6.25MHz      |
| K Counter clock( $Mf_0$ )       | 100MHz       |

### C. POST-PROCESSING CIRCUIT FOR THE MURO-TRNG ARCHITECTURE

To reduce the bias from the generated raw random bits from the MURO-TRNG architecture XOR corrector [22], [23] based post-processing circuit is used in this paper. XOR corrector based post-processing circuit is designed

by using 2 bit shift register circuit is shown in Figure.6. The proposed post-processing is applied on 2 consecutive bits to reduce the bias variation and it reduces throughput of MURO-TRNG architecture by 2. Sampling frequency is used as a clock frequency for the proposed post-processing circuit.

### III. EXPERIMENTAL RESULTS

Proposed MURO-TRNG architecture is designed by using VHDL (VHSIC HDL) and implemented on three Artix7, Kintex-7 and Zynq7000 FPGA (Field programmable Gate Array) board. It is synthesized and simulated by Xilinx Vivado2015.2 tool. FPGA system clock of the value 100MHz is used to generate ADPLL based ring oscillator circuits reference frequencies. Also ADPLL K counter and DCO clocks are generated by using FPGA system clock. By using FPGA system clock and divide by 2 counter circuits, three different reference frequencies (25MHz, 12.5MHz and 6.25MHz) of conventional ADPLL are generated. Conventional ADPLL output (IDout) is passed through divide by 2 counter circuit to generate sampling frequency. This sampling frequency is used as a clock frequency for 11DFFs and XOR Corrector based post-processing circuit in the proposed MURO-TRNG architecture. Simulation output waveform for the ADPLL based ring oscillator circuit implemented on Artix7 FPGA with reference frequency 100KHz is shown in Figure 7. In this figure, signal clk represents FPGA system clock, signal t represents T FF input, signal reset represents reset ports, fref represents 100KHz reference frequency, signal idout represents DCO output, signal fidclk represents clock with value  $Mf_0$  or  $2Nf_0$ , and signal fdco represents divide by  $N/2$  counter circuit output. Similarly other ADPLL based ring oscillator circuits are simulated in this work. In this paper, MURO-TRNG architectures designed using Artix7 FPGA simulation output waveforms are shown. Simulation output waveform for the proposed MURO-TRNG architecture with conventional ADPLL ( $f_0=25\text{MHz}$ ) is shown in Figure.8. In Figure.8, signal clk represents FPGA system clock, signal t represents T FF input, signal reset represents reset ports for MURO-TRNG architecture, signal xorgateop represents XOR gate output for MURO-TRNG architecture, signal fs25MHz represents reference signal for the conventional ADPLL, signal c1 represents sampling clock, signal randop represents raw random bits outputs for MURO-TRNG architecture and signal randproc



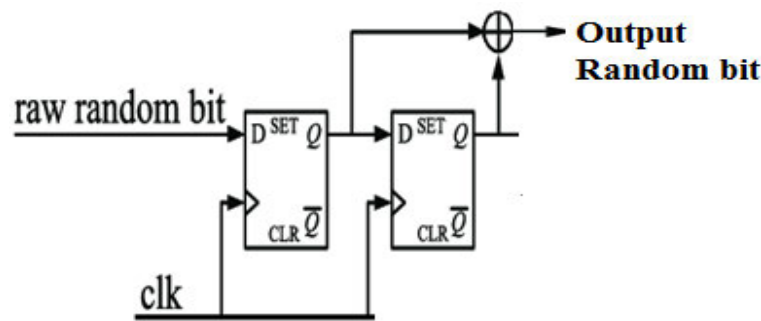


FIGURE 6. XOR corrector based post-processing circuit.

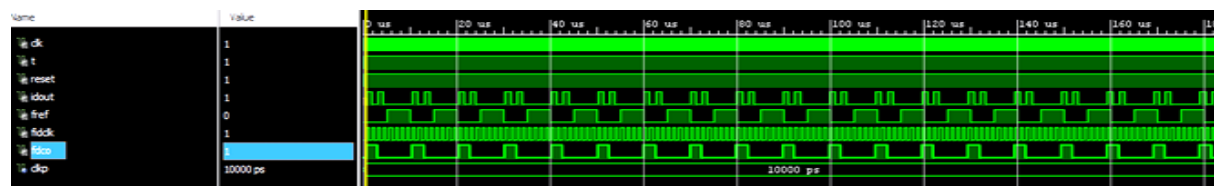


FIGURE 7. Simulation output waveform for the proposed ADPLL based ring oscillator circuit with reference frequency 100 KHz.

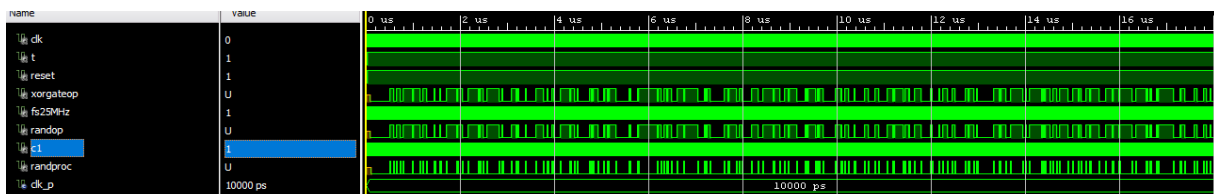


FIGURE 8. Simulation output waveform for the proposed MURO-TRNG architecture when conventional ADPLL reference frequency is 25MHz.

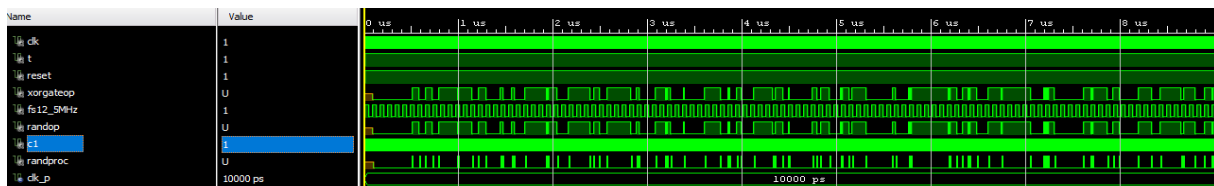


FIGURE 9. Simulation output waveform for the proposed MURO-TRNG architecture when conventional ADPLL reference frequency is 12.5MHz.

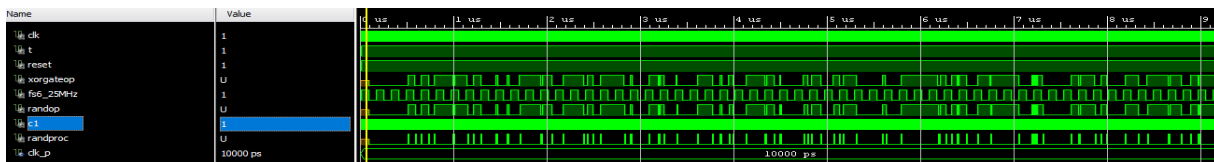


FIGURE 10. Simulation output waveform for the proposed MURO-TRNG architecture when conventional ADPLL reference frequency is 6.25MHz.

represents final output random bits after post-processing circuit in the proposed MURO-TRNG architecture. Simulation output waveform for the proposed MURO-TRNG architecture with conventional ADPLL ( $f_o=12.5\text{MHz}$ ) is

shown in Figure.9. In Figure.9, signal clk represents FPGA system clock, signal t represents T FF input, signal reset represents reset ports for MURO-TRNG architecture, signal xorgateop represents XOR gate output for MURO-TRNG

**TABLE 5.** NIST test result for MURO- TRNG architecture at  $F_0 = 25\text{MHz}$  (ARTIX7).

| NIST Test                         | P-value    | Result |
|-----------------------------------|------------|--------|
| Run                               | 0.1639     | Pass   |
| DFT                               | 0.1069     | Pass   |
| Serial                            | 0.9995     | Pass   |
| Frequency                         | 0.9900     | Pass   |
| Block Frequency                   | 0.7148     | Pass   |
| Cumulative sum (Forward)          | 0.05818    | Pass   |
| Cumulative sum (Backward)         | 0.05818    | Pass   |
| Random excursions                 | 0.0500     | Pass   |
| Non Overlapping Template Matching | 0.4524     | Pass   |
| Overlapping Template Matching     | 0.3972     | Pass   |
| Binary Matrix                     | 5.9797e-05 | Fail   |

**TABLE 6.** NIST test result for MURO- TRNG architecture at  $F_0 = 12.5\text{MHz}$  (ARTIX7).

| NIST Test                         | P-value   | Result |
|-----------------------------------|-----------|--------|
| Run                               | 0.0574    | Pass   |
| DFT                               | 0.1069    | Pass   |
| Serial                            | 0.9995    | Pass   |
| Frequency                         | 0.9900    | Pass   |
| Block Frequency                   | 0.7148    | Pass   |
| Cumulative sum (Forward)          | 0.01815   | Pass   |
| Cumulative sum (Backward)         | 0.01815   | Pass   |
| Random excursions                 | 0.0500    | Pass   |
| Non Overlapping Template Matching | 0.4524    | Pass   |
| Overlapping Template Matching     | 0.3972    | Pass   |
| Binary Matrix                     | 5.979e-05 | Fail   |

**TABLE 7.** NIST test result for MURO- TRNG architecture at  $F_0 = 6.25\text{MHz}$  (ARTIX7).

| NIST Test                         | P-value   | Result |
|-----------------------------------|-----------|--------|
| Run                               | 0.95886   | Pass   |
| DFT                               | 0.1069    | Pass   |
| Serial                            | 0.9995    | Pass   |
| Frequency                         | 0.9900    | Pass   |
| Block Frequency                   | 0.7149    | Pass   |
| Cumulative sum (Forward)          | 0.01306   | Pass   |
| Cumulative sum (Backward)         | 0.01306   | Pass   |
| Random excursions                 | 0.0500    | Pass   |
| Non Overlapping Template Matching | 0.4524    | Pass   |
| Overlapping Template Matching     | 0.3972    | Pass   |
| Binary Matrix                     | 5.979e-05 | Fail   |

architecture, signal fs12\_5MHz represents reference signal for the conventional ADPLL, signal c1 represents sampling clock, signal randop represents raw random bits outputs for MURO-TRNG architecture and signal randproc represents final output random bits after post-processing circuit in the proposed MURO-TRNG architecture. A simulation output waveform for the proposed MURO-TRNG architecture with conventional ADPLL ( $f_0=6.25\text{MHz}$ ) is shown in Figure.10. In Figure.10, signal clk represents FPGA system clock, signal t represents T FF input, signal reset represents reset ports for MURO-TRNG architecture, signal xorgateop represents XOR gate output for MURO-TRNG architecture, signal fs6\_25MHz represents reference signal for the conventional ADPLL, signal c1 represents sampling clock, signal randop represents raw random bits outputs for MURO-TRNG

architecture and signal randproc represents final output random bits after post-processing circuit in the proposed MURO-TRNG architecture. Generated output random bits from the post-processing circuit is stored in a text file at the rising edge of clock signal c1 for all MURO-TRNG architectures. Data path delay of 4.738ns yields a throughput value 211 Mbps [21] for the MURO-TRNG architecture implemented on Artix7 FPGA when the conventional ADPLL reference frequency is 25MHz. Data path delay of 4.594ns yields a throughput value 217.67 Mbps [21] for the MURO-TRNG architecture implemented on Artix7 FPGA when the conventional ADPLL reference frequency is 12.5MHz. Data path delay of 4.594ns yields a throughput value 217.67 Mbps [21] for the MURO-TRNG architecture implemented on Artix7 FPGA when the conventional ADPLL reference frequency

**TABLE 8.** Performance comparison among MURO-TRNG and existing trng architectures.

| Reference                                            | Entropy Source                    | Device                            | Hardware Resource                | Throughput                         | Post-Processing |
|------------------------------------------------------|-----------------------------------|-----------------------------------|----------------------------------|------------------------------------|-----------------|
| [26]                                                 | RS-Latch                          | Xilinx XC4VFX20                   | 580 Slices                       | 12.5 Mbps                          | No              |
| [27]                                                 | Metastability/Jitter              | Xilinx XCKU040                    | 1 PLL; 5 Primitives; 5 Slices    | 100 Mbps                           | Yes             |
| [28]                                                 | PLL jitter                        | Altera Stratix                    | 120 LE                           | > 1 Mbps                           | Yes             |
| [29]                                                 | Metastability                     | Virtex-5                          | n/a LUT 64 Latches               | 20 Mbps                            | No              |
| [30]                                                 | Free running ring oscillator      | Xilinx Spartan-6 FPGA             | 10 LUTs and 5 FFs                | 1.15 Mbps                          | No              |
| [31]                                                 | PLL                               | Xilinx FPGA                       |                                  | 6.25Mbps                           | No              |
| [32]                                                 | Free-running oscillator           | Xilinx XC2VP30                    | 973 Slices                       | 2.5 Mbps                           | No              |
| [33]                                                 | PLL/DLL                           | Altera APEX EP20K200EFC484-2X     | 121 LCs, 4 ESB and 1 PLL         | 69 Kbps                            | No              |
| [34]                                                 | Free-running oscillator           | Xilinx XC2VP30                    | 12LUTs and 24 registers          | 300 Kbps                           | No              |
| [35]                                                 | RO-PLL                            | Xilinx XC3S500E                   | 12 LUTs, 4 registers and 1 PLL   | 2Mbps                              | No              |
| [36]                                                 | MURO                              | Spartan 6 Cyclone V SmartFusion 2 | 521 LUTs<br>525 LUTs<br>545 LUTs | 2.57 Mbps<br>2.2 Mbps<br>3.62 Mbps | No              |
| [36]                                                 | Self timed ring (STR)             | Spartan 6 Cyclone V SmartFusion 2 | 346 LUTs<br>352 LUTs<br>350 LUTs | 154 Mbps<br>245 Mbps<br>188 Mbps   | No              |
| [37]                                                 | Ring oscillator                   | Spartan 3A                        | 528LUTs                          | 6 Mbps                             | No              |
| [40]                                                 | Lightweight hybrid entropy source | Xilinx Artix-7                    | 12 LUTs and 10 DFFs              | 150Mbps                            | No              |
| [41]                                                 | Metastability                     | Xilinx Zynq-7000 XC7Z020          | 38 LUTs and 121 DFFs             | 300 Mbps                           | Yes             |
| Proposed MURO-TRNG( $f_0=25\text{MHz}$ )Artix7       | ADPLL Jitter                      | Xilinx XC7A35T-CPG236-1           | 2 LUTs and 2 DFFs                | 211 Mbps                           | Yes             |
| Proposed MURO-TRNG( $f_0=12.5\text{MHz}$ ) Artix7    | ADPLL Jitter                      | Xilinx XC7A35T-CPG236-1           | 3 LUTs and 3 DFFs                | 217.67 Mbps                        | Yes             |
| Proposed MURO-TRNG( $f_0=6.25\text{MHz}$ ) Artix7    | ADPLL Jitter                      | Xilinx XC7A35T-CPG236-1           | 4 LUTs and 4 DFFs                | 217.67 Mbps                        | Yes             |
| Proposed MURO-TRNG( $f_0=25\text{MHz}$ )Kintex-7     | ADPLL Jitter                      | Xilinx 7k70t-fbv484               | 2 LUTs and 2 DFFs                | 253.48 Mbps                        | Yes             |
| Proposed MURO-TRNG( $f_0=12.5\text{MHz}$ ) Kintex-7  | ADPLL Jitter                      | Xilinx 7k70t-fbv484               | 3 LUTs and 3 DFFs                | 260.07 Mbps                        | Yes             |
| Proposed MURO-TRNG( $f_0=6.25\text{MHz}$ ) Kintex-7  | ADPLL Jitter                      | Xilinx 7k70t-fbv484               | 4 LUTs and 4 DFFs                | 260.07 Mbps                        | Yes             |
| Proposed MURO-TRNG( $f_0=25\text{MHz}$ ) Zynq 7000   | ADPLL Jitter                      | Xilinx 7z010-clg225               | 2 LUTs and 2 DFFs                | 207.85 Mbps                        | Yes             |
| Proposed MURO-TRNG( $f_0=12.5\text{MHz}$ ) Zynq 7000 | ADPLL Jitter                      | Xilinx 7z010-clg225               | 3 LUTs and 3 DFFs                | 209.68 Mbps                        | Yes             |
| Proposed MURO-TRNG( $f_0=6.25\text{MHz}$ ) Zynq 7000 | ADPLL Jitter                      | Xilinx 7z010-clg225               | 4 LUTs and 4 DFFs                | 206.82 Mbps                        | Yes             |

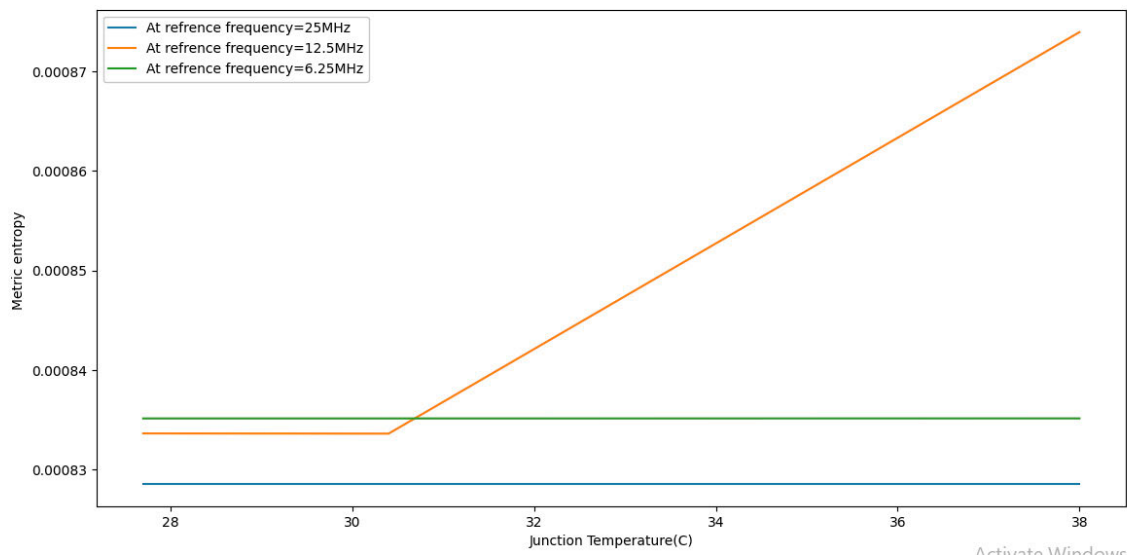
is 6.25MHz. Data path delay of 3.945ns yields a throughput value 253.48 Mbps [21] for the MURO-TRNG architecture implemented on Kintex-7 FPGA when the conventional ADPLL reference frequency is 25MHz. Data path delay of 3.845ns yields a throughput value 260.07 Mbps [21] for the MURO-TRNG architecture implemented on Kintex-7 FPGA when the conventional ADPLL reference frequency is 12.5MHz. Data path delay of 3.845ns yields a throughput value 260.07 Mbps [21] for the MURO-TRNG architecture implemented on Kintex-7 FPGA when the conventional ADPLL reference frequency is 6.25MHz. Data path delay

of 4.811ns yields a throughput value 207.85 Mbps [21] for the MURO-TRNG architecture implemented on Zynq7000 FPGA when the conventional ADPLL reference frequency is 25MHz. Data path delay of 4.769ns yields a throughput value 209.68Mbps [21] for the MURO-TRNG architecture implemented on Zynq7000 FPGA when the conventional ADPLL reference frequency is 12.5MHz. Data path delay of 4.835ns yields a throughput value 206.82 Mbps [21] for the MURO-TRNG architecture implemented on Zynq7000 FPGA when the conventional ADPLL reference frequency is 6.25MHz.



**TABLE 9.** Power and Energy/bit results comparison among MURO-TRNG and existing TRNG architectures.

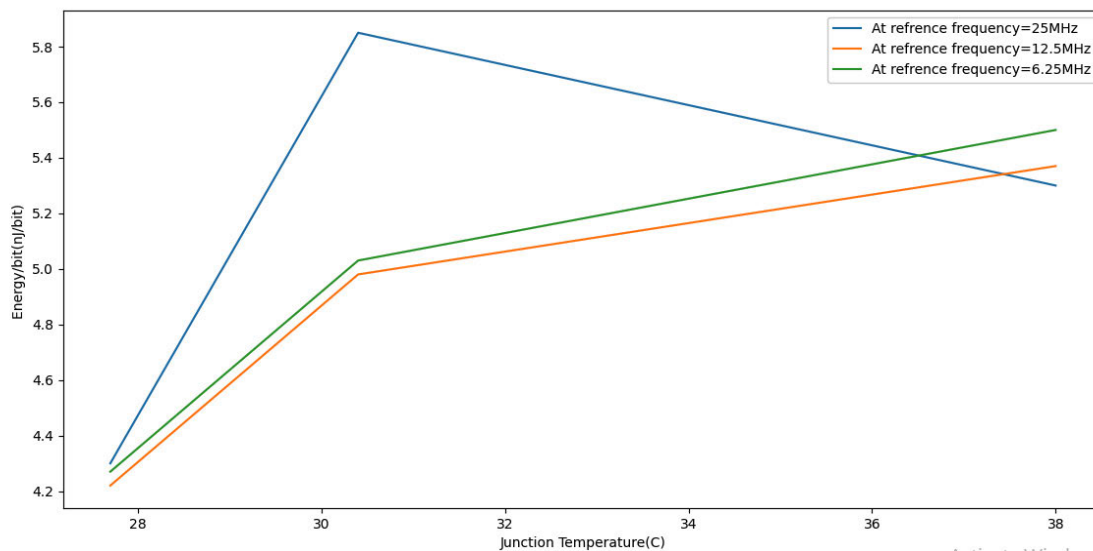
| Parameter                                         | [38]         | [39]       | Proposed MURO-TRNG( $f_o=25\text{MHz}$ ) | Proposed MURO-TRNG( $f_o=12.5\text{MHz}$ ) | Proposed MURO-TRNG( $f_o=6.25\text{MHz}$ ) |
|---------------------------------------------------|--------------|------------|------------------------------------------|--------------------------------------------|--------------------------------------------|
| Power Consumption(Artix7)                         |              |            | 1.073 W                                  | 1.085 W                                    | 1.097 W                                    |
| Power Consumption(Kintex-7)                       |              |            | 1.092 W                                  | 1.099 W                                    | 1.112 W                                    |
| Power Consumption(Zynq 7000)                      |              |            | 1.116 W                                  | 1.126 W                                    | 1.138 W                                    |
| Energy/bit(Artix7 FPGA)                           |              |            | 5.85 nJ/bit                              | 4.98 nJ/bit                                | 5.03 nJ/bit                                |
| Energy/bit(Kintex-7 FPGA)                         |              |            | 4.3 nJ/bit                               | 4.22 nJ/bit                                | 4.27 nJ/bit                                |
| Energy/bit(Zynq7000 FPGA)                         |              |            | 5.3 nJ/bit                               | 5.37 nJ/bit                                | 5.5 nJ/bit                                 |
| Power Consumption(Cyclone IV EEP4CE115F29CN) [38] | 0.04126 W    |            |                                          |                                            |                                            |
| Power Consumption (Virtex5)[39]                   |              | 0.238 W    |                                          |                                            |                                            |
| Energy/bit(Cyclone IV EEP4CE115F29CN)[38]         | 11.78 nJ/bit |            |                                          |                                            |                                            |
| Energy/bit(Virtex5) [39]                          |              | 568 nJ/bit |                                          |                                            |                                            |

**FIGURE 11.** Results of voltages and temperatures tests with Metric Entropy.

MATLAB R2020a tool is used to conduct NIST (National Institute of Standards and Technology) SP 800-22 test for validating the generated random bits. MATLAB R2020a tool read the text file and if sequence of bits with a P-value  $\geq$  of 0.001 passes the NIST test [24], [25]. A total of 2000 random bits are considered to conduct the NIST test. In this work, NIST test is conducted for three reference frequencies of conventional ADPLL  $f_o=25\text{MHz}$ ,  $f_o=12.5\text{MHz}$  and  $f_o=6.25\text{MHz}$ . The NIST Test Result for MURO- TRNG Architecture when conventional ADPLL reference frequency is 25MHz and it is implemented on Artix7 is shown in Table 5. NIST Test Result for MURO- TRNG Architecture when

conventional ADPLL reference frequency is 12.5MHz and it is implemented on Artix7 is shown in Table 6. NIST Test Result for MURO- TRNG Architecture when conventional ADPLL reference frequency is 6.25MHz and it is implemented on Artix7 is shown in Table 7. These Tables show that generated output bit stream from the proposed TRNG architectures is purely random.

Performance and Power and Energy/bit results comparison among MURO-TRNG and existing TRNG architectures are shown in Table 8 and Table 9 respectively. Our proposed MURO-TRNG architecture implemented on three different FPGA boards with three different reference frequencies of



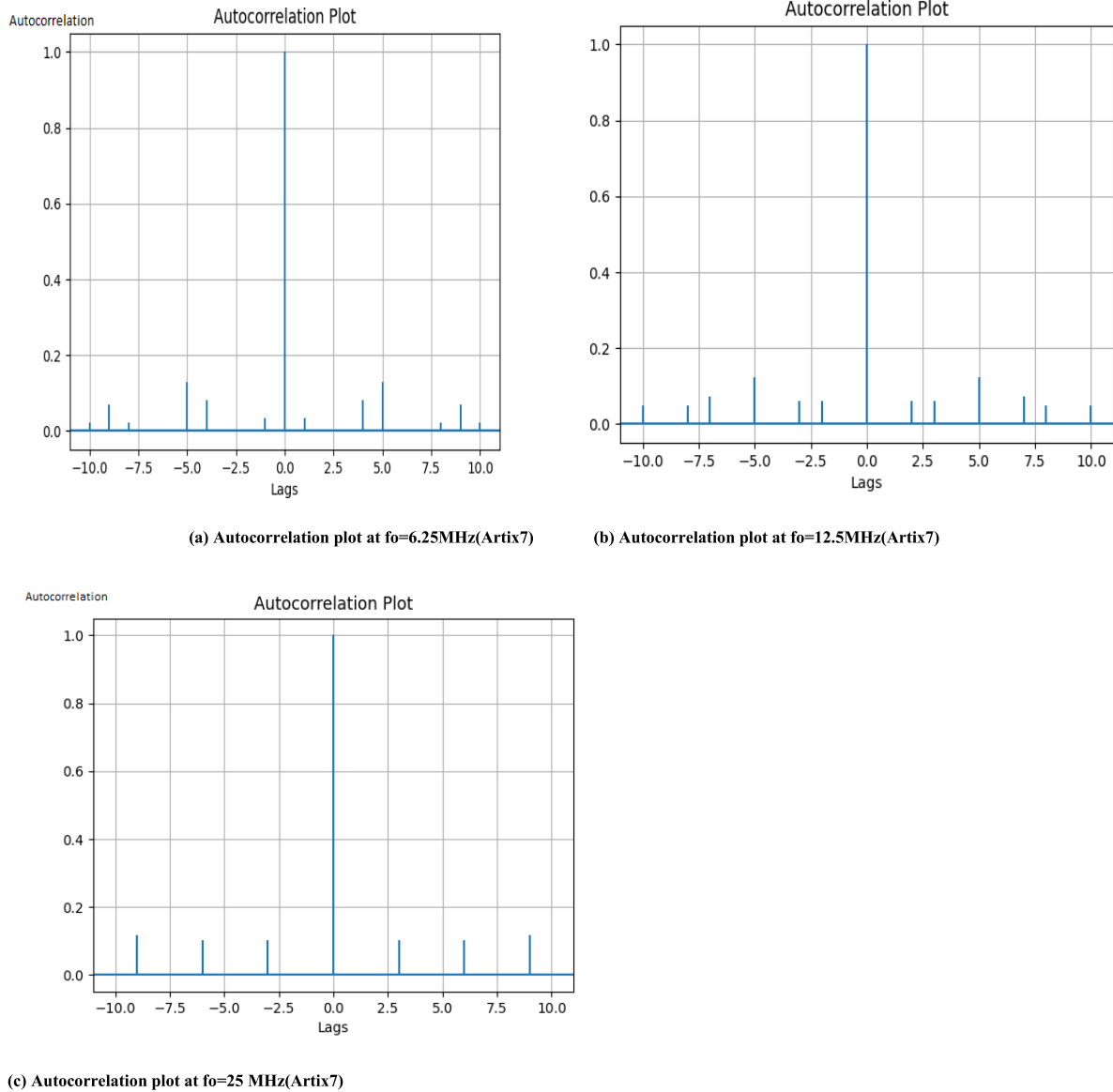
**FIGURE 12.** Results of voltages and temperatures tests with Energy/bit.

conventional ADPLL consumes LUTs in the range of 2-4 and D FFs in the range of 2-4. The throughput value of the Our proposed MURO-TRNG architecture implemented on three different FPGA boards with three different reference frequencies of conventional ADPLL is in the range of 206.82 Mbps-260.07 Mbps. Paper [28] uses PLL jitter as an entropy source but this design has smaller throughput value in comparison with proposed MURO-TRNG design.

Paper [26], Paper [27], Paper [29], Paper [30], Paper [31], Paper [32], Paper [33], Paper [34], Paper [35], Paper [37] and Paper [40] has smaller throughput value than proposed design. In paper [36], MURO TRNG architectures are implemented on different devices but they utilize more hardware resources in comparison with proposed MURO-TRNG architecture. Paper [37] uses ring oscillator as entropy source for the TRNG architecture and they are implemented on Spartan 3A FPGA. This TRNG architecture consumes 528LUTs and throughput value for this design is 6 Mbps. Paper [41] consume larger hardware resources in comparison with proposed TRNG design.

Total on chip power consumption of the proposed MURO-TRNG architecture implemented on three different FPGA boards with three different reference frequencies of conventional ADPLL is in the range of 1.073 W to 1.138 W. Energy/bit values of the proposed MURO-TRNG architecture implemented on three different FPGA boards with three different reference frequencies of conventional ADPLL is in the range of 4.22 nJ/bit to 5.85 nJ/bit. Paper [38] and Paper [39] has higher energy/bit value in comparison with the proposed MURO-TRNG architectures. To evaluate the reliability of the proposed MURO-TRNG architectures, three FPGA boards are considered. The Voltage range for the Artix7 FPGA is 0.95V to 1.05V, and its operating

Temperature range is from 0 to 85 degree Celsius. The Voltage range for Kintex-7 FPGA is 0.97V to 1.03V, and its operating Temperature range is from 0 to 85 degree Celsius. The Voltage range for Zynq7000 FPGA is 0.95V to 1.05V, and its operating Temperature range is from 0 to 85 degree Celsius. From the synthesis results, junction temperature values of Artix7 (30.4 degrees Celsius), Kintex-7 (27.7 degrees Celsius), and Zynq7000 (38 degrees Celsius) are found, and these values are used to test the voltage and temperature variations of the proposed TRNG design. Shannon entropy divided by bitstream length [7](Metric Entropy) is calculated for the MURO-TRNG architecture implemented on three different FPGA boards with three different reference frequencies of conventional ADPLL by using generated 2000 random bits. Figure 11 proves that proposed MURO-TRNG architecture output random bits are uniform and they are not affected with junction temperature and voltage changes. Energy/bit(Total on chip Power consumption/throughput) variations with junction temperature and voltages are show in Figure 12. This Figures shows that proposed MURO-TRNG architectures are energy efficient and uniform with junction temperature and voltages changes. Autocorrelation function (ACF) is calculated by using generated 2000 output bits to check the randomness of the proposed MURO-TRNG architectures. Figure 13 shows the Autocorrelation function plots at three different reference frequencies of conventional ADPLL implemented on Artix7 FPGA. These Figures show independence of generated output bits from the proposed MURO-TRNG architectures, as most of the ACF [42] factors lie between 0 to 0.1. From Figure 13, it is clear that generated binary sequence from the proposed TRNG architectures are more resistant to correlation analysis attacks.



**FIGURE 13.** (a) Autocorrelation plot at  $f_o=6.25\text{MHz}$ (Artix7) (b) Autocorrelation plot at  $f_o=12.5\text{MHz}$ (Artix7) (c) Autocorrelation plot at  $f_o=25\text{ MHz}$ (Artix7).

#### IV. CONCLUSION

In the proposed technique, by utilizing timing jitter produced by ADPLL based ring oscillators circuit, highly secure and trustworthy TRNG is produced. Ring oscillators designed with ADPLL are almost in locked condition, and their DCO output has higher frequencies compared with the reference frequencies. From the simulation output waveform for the proposed MURO-TRNG architecture, it is clear that produced output random bits from the post-processing circuit are able to reduce bias from the raw random bits. Implemented on Kintex-7 FPGA at the reference frequency of 6.25MHz has higher throughput value in comparison with other MURO-TRNG architectures. Proposed TRNG

architectures implemented at 25MHz reference frequency consume only 2LUTs and 2 DFFs, and their area consumption is the lowest among implemented TRNG architectures. The TRNG architecture implemented on Kintex-7 at reference frequency of 12.5 MHz has an Energy dissipation per bit value is 2.79 times lower than paper [38]. TRNG architecture implemented on Artix7 at reference frequency of 25 MHz has lowest power consumption value in comparison with other proposed TRNG architectures. Output random bits produced from the post-processing circuit are stored in a text file at the rising edge of sampling clock frequency. MATLAB R2020a tool conducts the NIST test and it validates the randomness of the MURO-TRNG design. From Table 8,

it is clear that proposed MURO-TRNG design consume less hardware and has a high throughput value in comparison with the existing TRNG design. Also from Table 9, it is clear that proposed designs are more energy efficient in comparison with existing literature. Proposed TRNG architectures can be used for a wide range of operating conditions. Designed and implemented MURO-TRNG architecture can be used in many applications such as cyber Security, IoT, wireless communication etc.

## ACRONYMS

|           |                                                |
|-----------|------------------------------------------------|
| ADPLL     | All Digital Phase Locked Loop.                 |
| TRNG      | True random number generator.                  |
| MURO-TRNG | Multiple Ring oscillator TRNG.                 |
| DCO       | Digital Controlled Oscillator.                 |
| PRNG      | Pseudo random number generator.                |
| PLL       | Phase locked loop.                             |
| NIST      | National Institute of standard and technology. |
| CSRO      | Current starve based ring oscillator.          |
| RO        | Ring oscillator.                               |
| LUT       | Look up table.                                 |
| FFs       | Flip-flops.                                    |
| DFFs      | D Flip-flops.                                  |
| VHDL      | VHSIC HDL.                                     |
| FPGA      | Field Programmable Gate Array.                 |
| ACF       | Autocorrelation function.                      |

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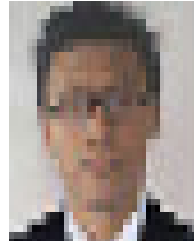
## DECLARATIONS COMPETING INTERESTS

The authors declare no competing interests.

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